HOFFT: Modeling **H**igh **O**rder Phases Using NU**FFT** Kernels

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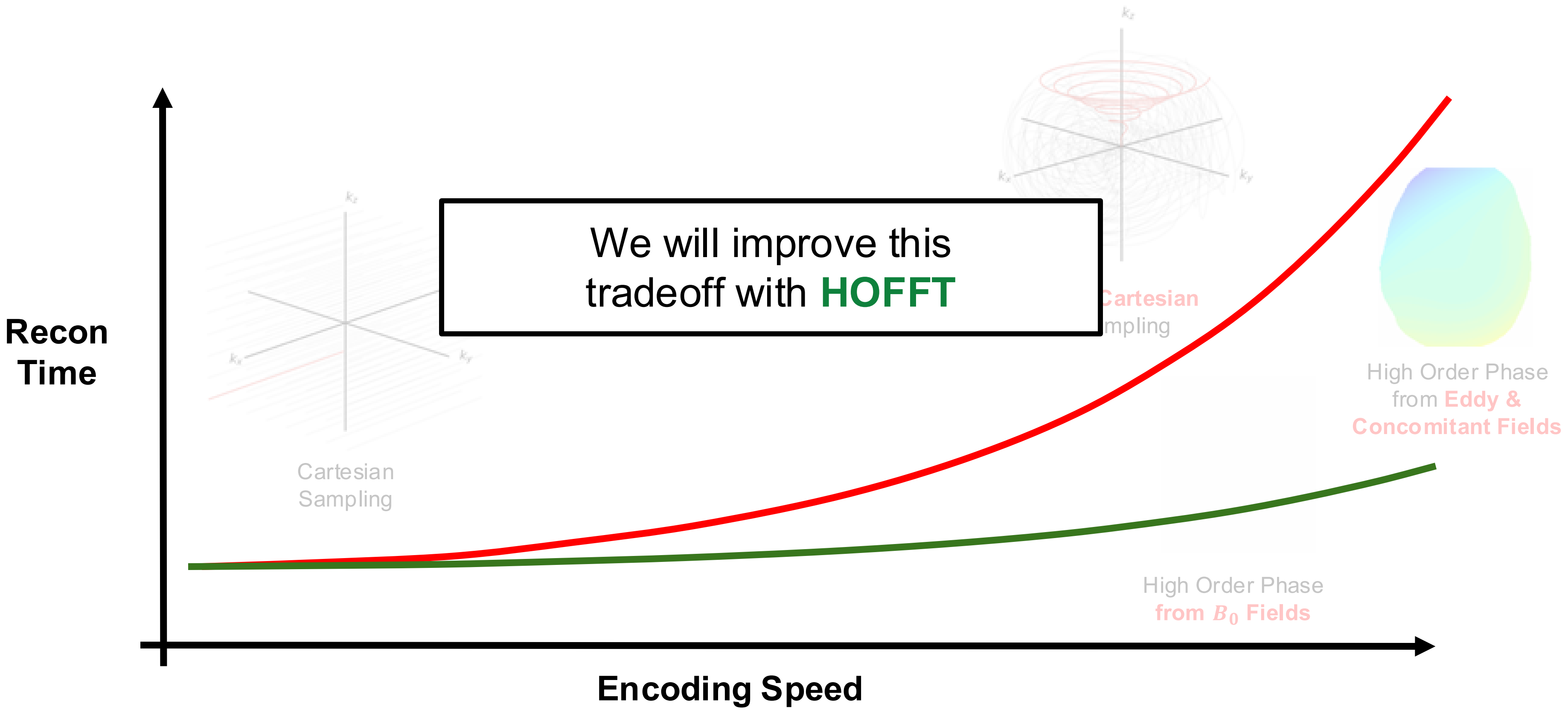
Declaration of Financial Interests or Relationships

Speaker Name: **Daniel Raz Abraham**

I have no financial interests or relationships to disclose with regard to the subject matter of this presentation.

Motivation

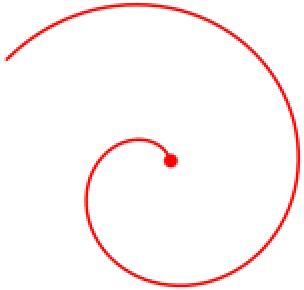
Efficient encoding schemes often require **expensive reconstructions**



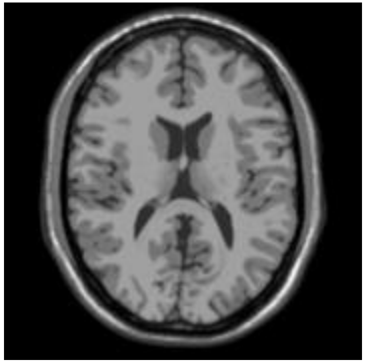
What is Higher Order Phase?

$$y(t) = \int_{\mathbf{r}} x(\mathbf{r}) e^{-j2\pi \mathbf{k}(t)\cdot\mathbf{r}} d\mathbf{r}$$

non-Cartesian MRI Model



$$= \int_{\mathbf{r}}$$



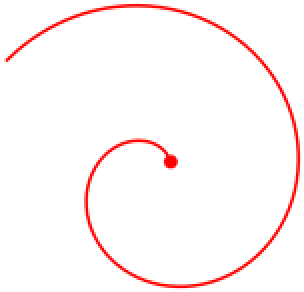
$$d\mathbf{r}$$

What is Higher Order Phase?

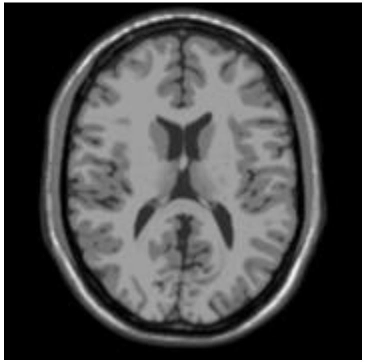
$$y(t) = \int_{\mathbf{r}} x(\mathbf{r}) e^{-j2\pi \mathbf{k}(t)\cdot\mathbf{r}} e^{-j2\pi \boldsymbol{\phi}(\mathbf{r})\boldsymbol{\alpha}(t)} d\mathbf{r}$$

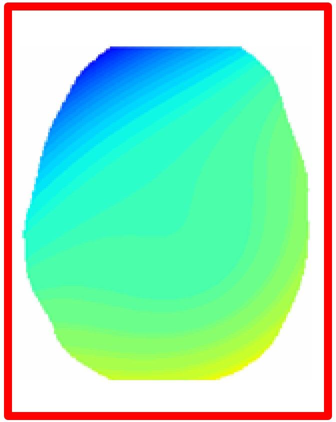
Spatio-temporal phase that is
spatially non-linear

non-Cartesian MRI Model with High Order Phase



$$= \int_{\mathbf{r}}$$





$$d\mathbf{r}$$

Existing Methods

$$y(t) = \int_{\mathbf{r}} x(\mathbf{r}) e^{-j2\pi \mathbf{k}(t)\cdot\mathbf{r}} e^{-j2\pi \phi(\mathbf{r}) \alpha(t)} d\mathbf{r}$$

$$\xrightarrow{\text{Discretization}} \quad \mathbf{y} \approx \mathbf{A}\mathbf{x}$$

Existing Techniques:

- Expanded encoding[1]
- Multi frequency interpolation[2]
- Time segmentation[3]
- Singular Vector Separation[4] -- **gold standard**

These either have poor computational complexity, large errors, or don't fully leverage the NUFFT operator

[1] Wilm … MRM 2018
[2] Man … MRM 1997
[3] Sutton … IEEE 2003
[4] Wilm … IEEE 2012

How To Use HOFFT

$$y(t) = \int_{\mathbf{r}} x(\mathbf{r}) e^{-j2\pi \mathbf{k}(t)\cdot \mathbf{r}} e^{-j2\pi \boldsymbol{\phi}(\mathbf{r})\boldsymbol{\alpha}(t)} \, d\mathbf{r}$$

You give us a description of your **high order phase**

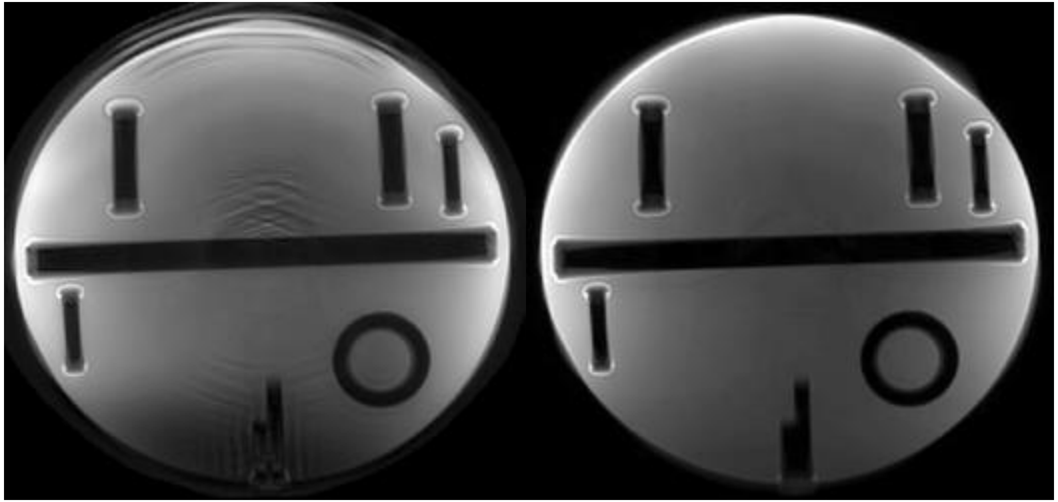
$$\mathbf{y} \approx \mathbf{A}_{\mathrm{HOFFT}} \mathbf{x}$$

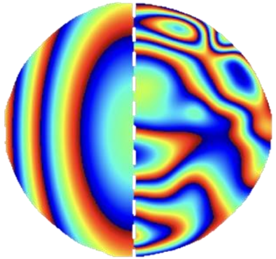
We will give you an **efficient forward model** with minimal approximation error

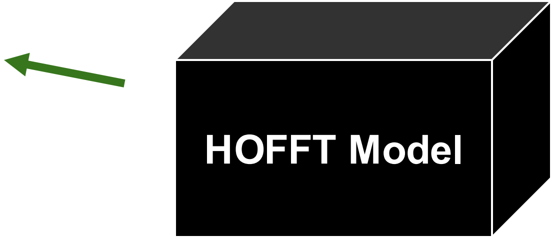
How HOFFT Performs

$$y(t) = \int_{\mathbf{r}} x(\mathbf{r}) e^{-j2\pi \mathbf{k}(t)\cdot\mathbf{r}} e^{-j2\pi \boldsymbol{\phi}(\mathbf{r})\boldsymbol{\alpha}(t)} d\mathbf{r}$$

Gold Standard (Equivalent Recon Time) | **HOFFT (Equivalent Recon Time)**



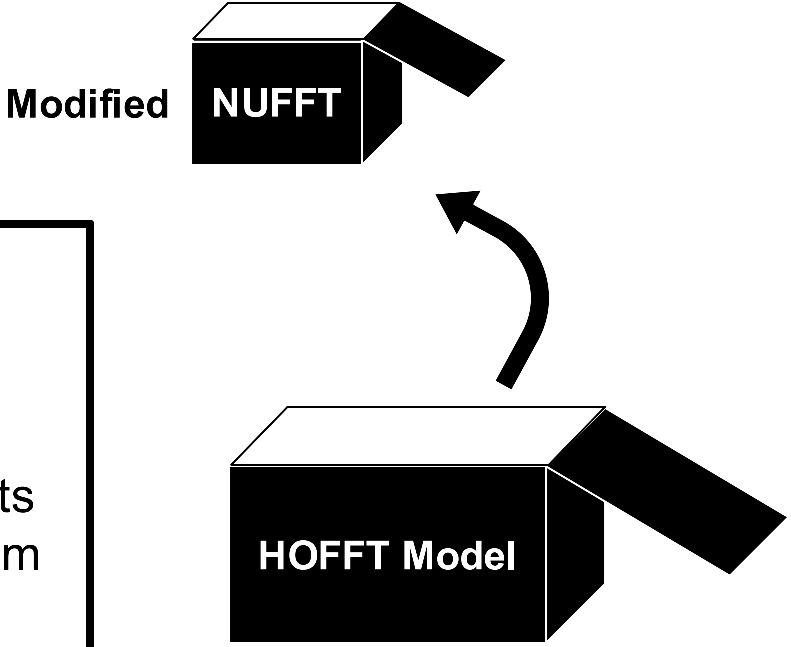




B_0 & Concomitant Field
0.8mm Spiral Example

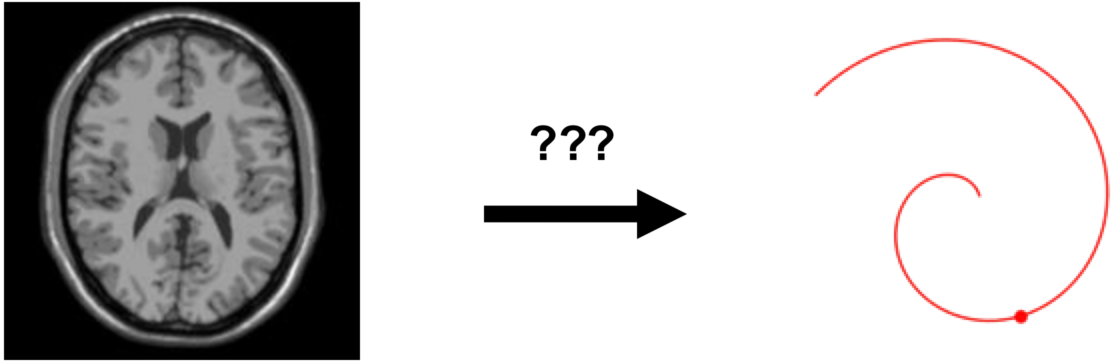
How HOFFT Works

NUFFTs weren’t designed for high order phase, HOFFT fixes that. **How?**



HOFFT Objectives

1. How does the NUFFT algorithm work?
2. Reformulate optimal NUFFT kernel weights and apodization as an optimization problem
3. Extending NUFFT to High Order Phase (HOFFT)



HOFFT Objectives

1. **How does the NUFFT algorithm work?**

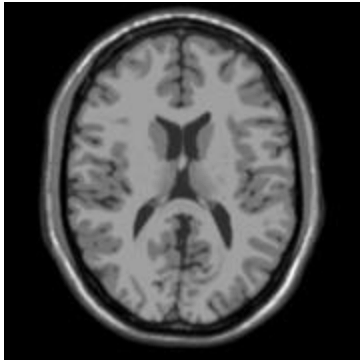
2. Reformulate optimal NUFFT kernel weights and apodization as an optimization problem

3. Extending NUFFT to High Order Phase (HOFFT)

Fessler … IEEE 2003, Beatty … MRM 2005

NUFFTs: Signal Processing Description (3x3 Example)

Given some Image



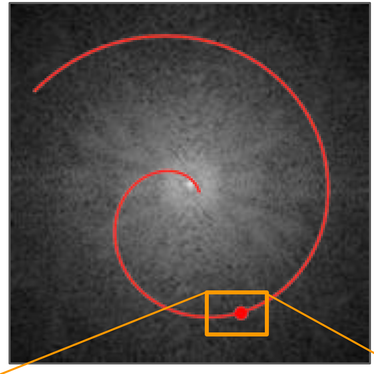
$x(\mathbf{r})$

×

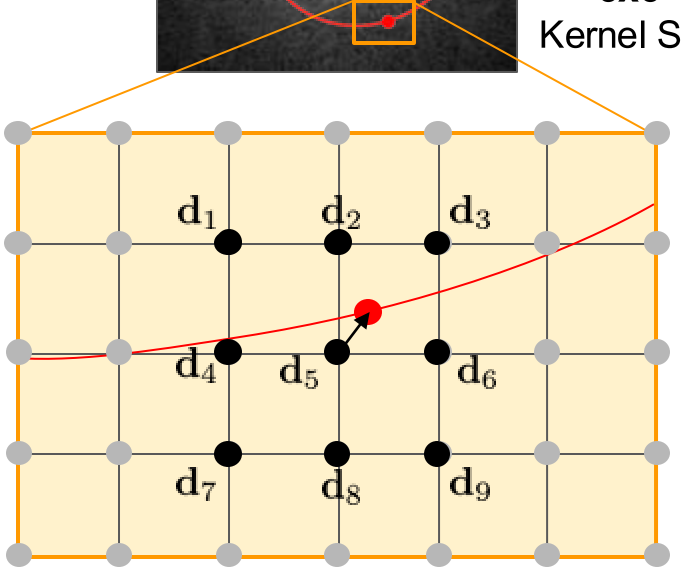
$b(\mathbf{r})$

FFT

We want non-Cartesian k-space



3x3
Kernel Size



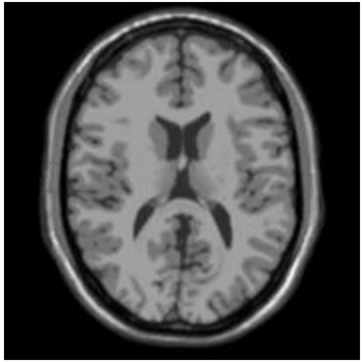
$$w_i(\mathbf{k}_{\text{dev}}) = \text{Kaiser-Bessel}(\mathbf{k}_{\text{dev}} + \mathbf{d}_i)$$

$$b(\mathbf{r}) = (\mathcal{F}\{\text{Kaiser-Bessel}(\mathbf{k})\})^{-1}$$

$$\sum_i w_i(\mathbf{k}_{\text{dev}}) \times \bullet_i \approx \bullet$$

NUFFTs: Linear Synthesis Description (3x3 Example)

Given some Image



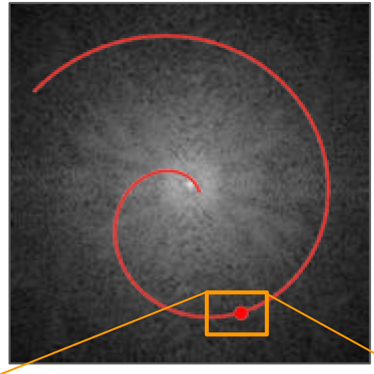
$x(\mathbf{r})$

$\times$

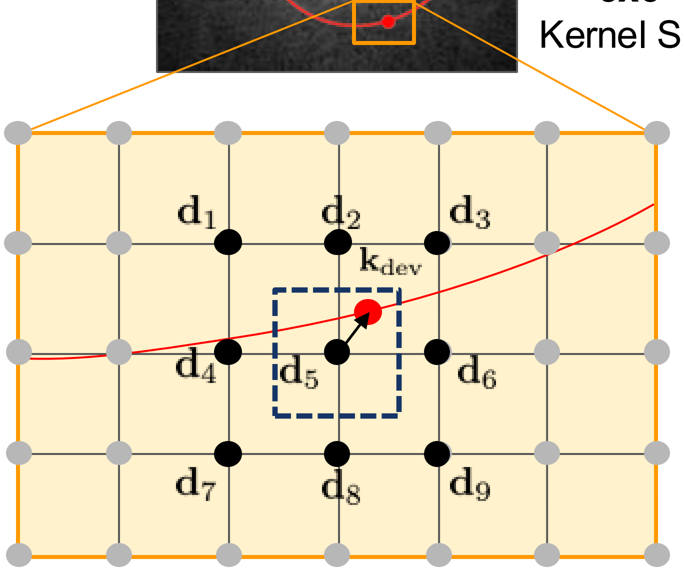
$b(\mathbf{r})$

FFT

We want non-Cartesian k-space



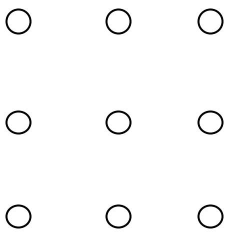
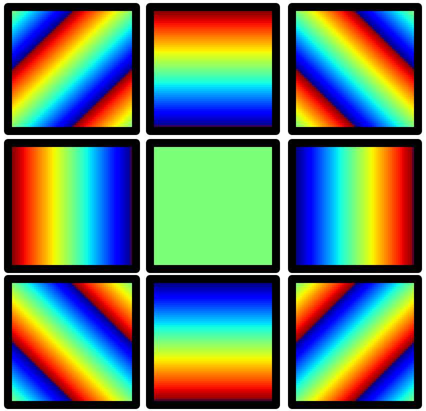
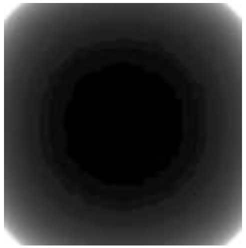
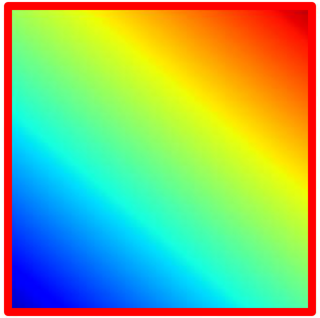
3x3
Kernel Size



K-space Shift Invariance

$$\sum_i w_i(\mathbf{k}_{\text{dev}}) \times \bullet_i \approx \textcolor{red}{\bullet}$$

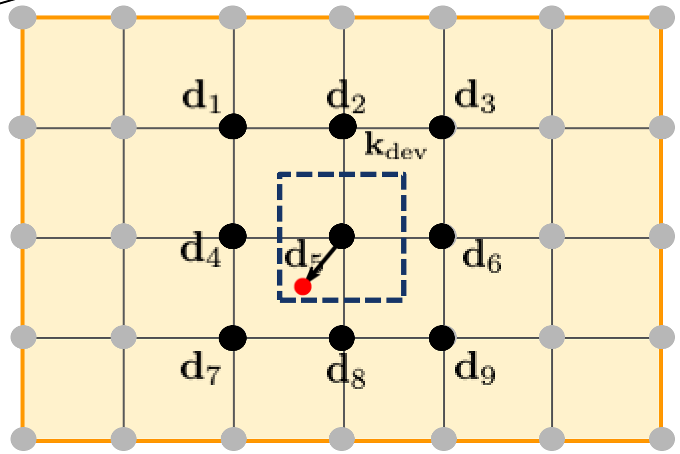
NUFFTs: Linear Synthesis Description (3x3 Example)

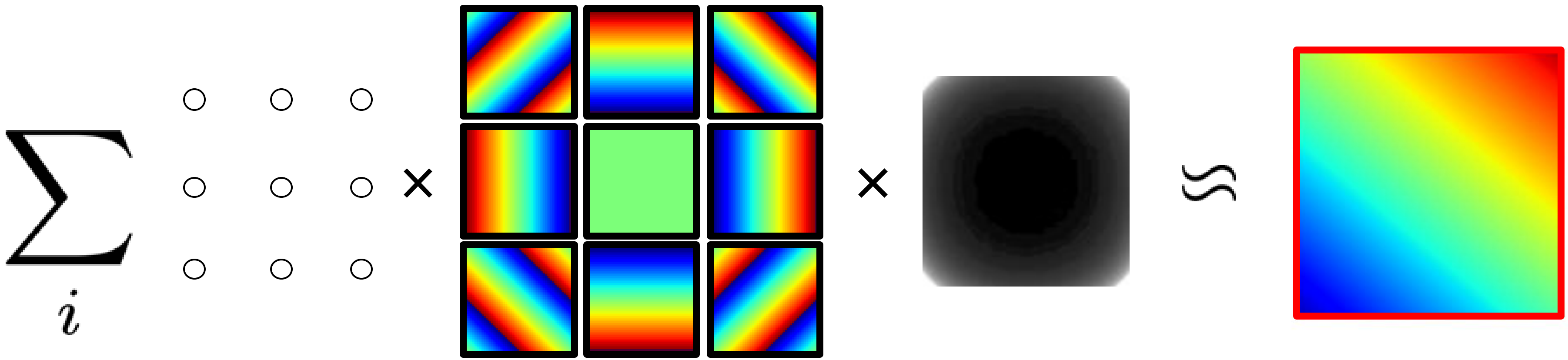
$\sum_i$  $\times$  $\times$  $\approx$ 

$$\sum_i w_i(\mathbf{k}_{\text{dev}})e^{-j2\pi\mathbf{d}_i\cdot\mathbf{r}}b(\mathbf{r}) \approx {\color{red}e^{-j2\pi\mathbf{k}_{\text{dev}}\cdot\mathbf{r}}}$$

K-space Shift Invariance

$$\sum_i w_i(\mathbf{k}_{\text{dev}}) \times \bullet_i \approx {\color{red}\bullet}$$





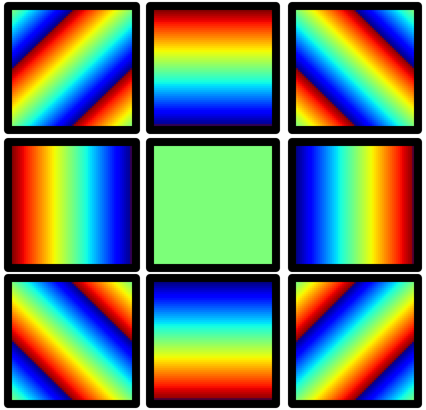
HOFFT Objectives

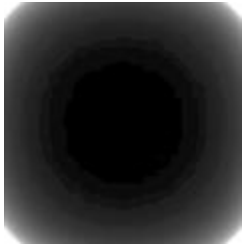
1. How does the NUFFT algorithm work?
2. **Reformulate optimal NUFFT kernel weights and apodization as an optimization problem**
3. Extending NUFFT to High Order Phase (HOFFT)

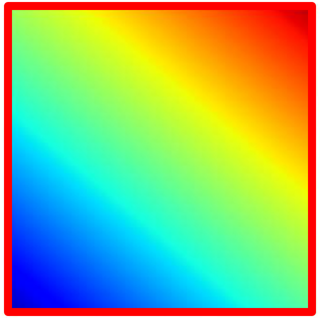
Optimal Kernel Weights & Apodization

Optimize   *Optimize*

$$\sum_i \; [w_i(\mathbf{k}_{\mathrm{dev}})] \times [e^{-j2\pi \mathbf{d}_i \cdot \mathbf{r}}] \times [b(\mathbf{r})] \approx [e^{-j2\pi \mathbf{k}_{\mathrm{dev}} \cdot \mathbf{r}}]$$







$$\min_{\mathbf{W},\mathbf{B}} || \sum_i \mathbf{W}_i \mathbf{E}_i \mathbf{B} - \mathbf{\Phi} ||_F^2$$

Non-Convex

Optimal Kernel Weights & Apodization: Algorithm

Approach 1
Alternating Least Squares

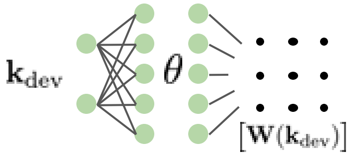
$$\mathbf{W}^{k+1} = \underset{\mathbf{W}}{\operatorname{argmin}} \, || \sum_i \mathbf{W}_i \mathbf{E}_i \mathbf{B}^{(k)} - \mathbf{\Phi} ||_F^2$$

$$\mathbf{B}^{k+1} = \underset{\mathbf{B}}{\operatorname{argmin}} \, || \sum_i \mathbf{W}_i^{(k+1)} \mathbf{E}_i \mathbf{B} - \mathbf{\Phi} ||_F^2$$

+ Fast on small-sized problems
- Lots of memory

Approach 2
Stochastic Gradient Descent + MLP

$$\min_{\theta, \mathbf{B}} || \sum_i \mathbf{E}_i \mathbf{W}_i(\theta) \mathbf{B} - \mathbf{\Phi} ||_F^2$$



+ Very memory efficient
- SGD can be slow

$$\min_{\mathbf{W}, \mathbf{B}} || \sum_i \mathbf{W}_i \mathbf{E}_i \mathbf{B} - \mathbf{\Phi} ||_F^2$$

Non-Convex

Joint Optimization vs KB NUFFT

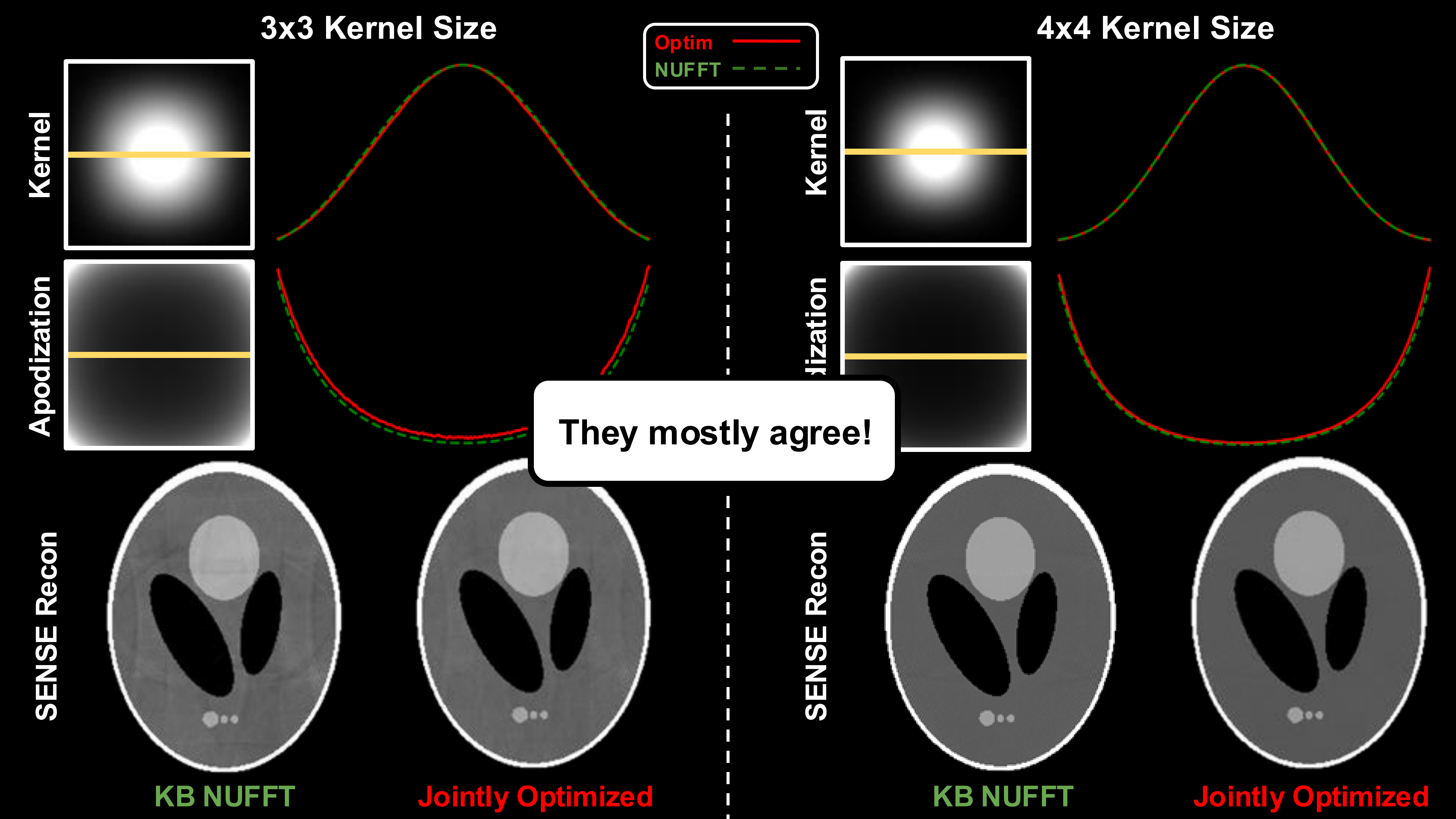
Signal processing approach:

$$w_i(\mathbf{k}_{\text{dev}}) = \text{Kaiser-Bessel}(\mathbf{k}_{\text{dev}} + \mathbf{d}_i)$$
$$b(\mathbf{r}) = (\mathcal{F}\{\text{Kaiser-Bessel}(\mathbf{k})\})^{-1}$$

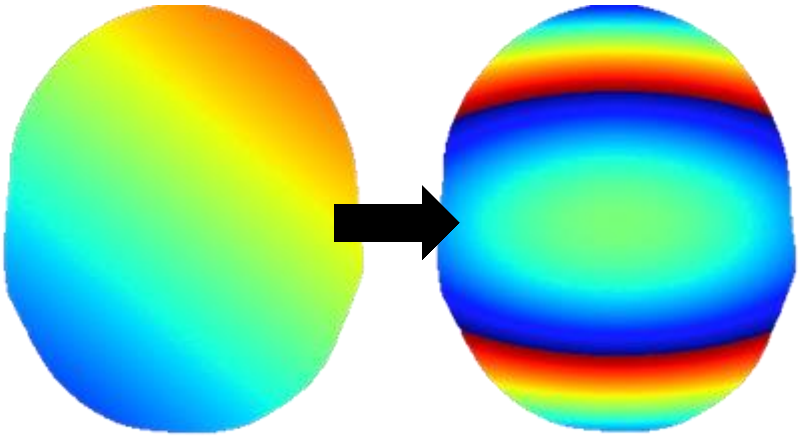
Linear algebra approach:

$$\mathbf{w}(\mathbf{k}_{\text{dev}}), b(\mathbf{r}) = \underset{\mathbf{W},\mathbf{B}}{\operatorname{argmin}} \left|\left| \sum_i \mathbf{W}_i \mathbf{E}_i \mathbf{B} - \mathbf{\Phi} \right|\right|_F^2$$

Will they agree?



3x3 Kernel Size
4x4 Kernel Size
Optim
NUFFT
Kernel
Apodization
SENSE Recon
Kernel
SENSE Recon
They mostly agree!
KB NUFFT
Jointly Optimized
KB NUFFT
Jointly Optimized



HOFFT Objectives

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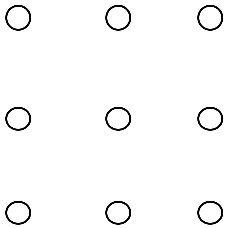
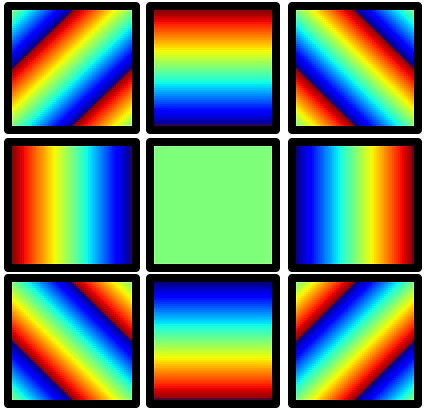
3. **Extending NUFFT to High Order Phase (HOFFT)**

NUFFT With High Order Phase

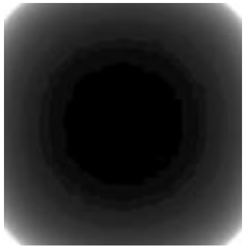
$$\sum_i$$



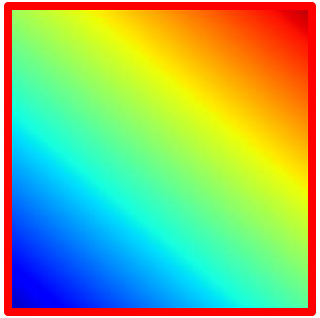
$$\times$$



$$\times$$



$$\approx$$



$$\left[w_i(\mathbf{k}_{\mathrm{dev}})\right] \quad \left[e^{-j2\pi \mathbf{d}_i \cdot \mathbf{r}}\right] \quad \left[b(\mathbf{r})\right] \quad \left[e^{-j2\pi \mathbf{k}_{\mathrm{dev}} \cdot \mathbf{r}}\right]$$

Replace with general **high order** phase

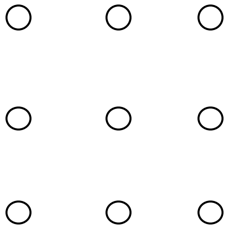
Recall the NUFFT model

NUFFT With High Order Phase

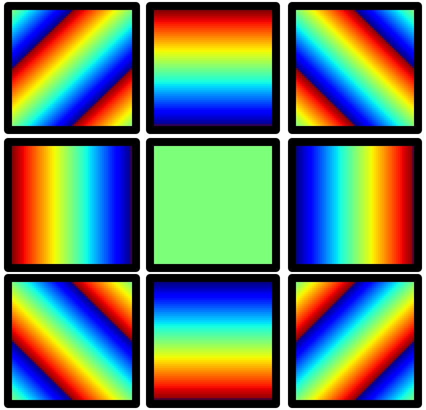
Optimize

Optimize

$$\sum_i$$



$\times$



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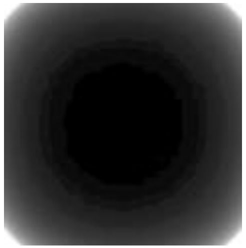


$\approx$

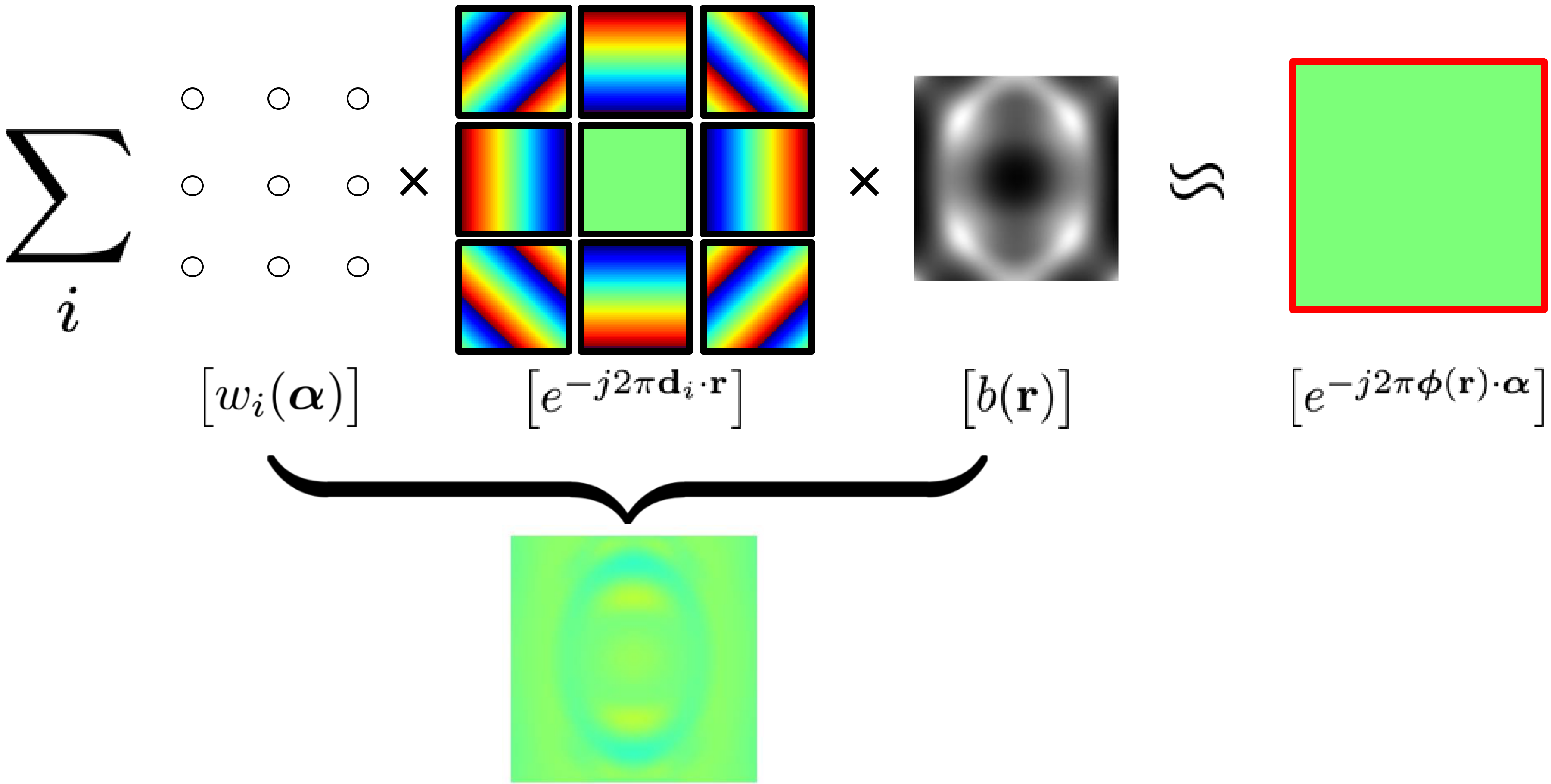
$[w_i(\boldsymbol{\alpha})]$ $[e^{-j2\pi \mathbf{d}_i \cdot \mathbf{r}}]$ $[b(\mathbf{r})]$ $[e^{-j2\pi \boldsymbol{\phi}(\mathbf{r}) \cdot \boldsymbol{\alpha}}]$

Replace with general **high order** phase

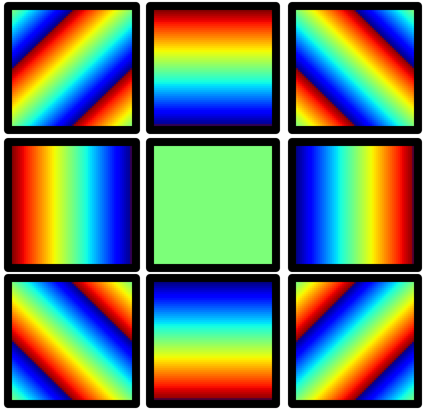
$$\min_{\mathbf{W},\mathbf{B}} || \sum_i \mathbf{W}_i \mathbf{E}_i \mathbf{B} - \mathbf{\Phi} ||_F^2$$

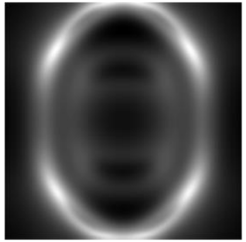
Almost identical optimization … how well does it work?

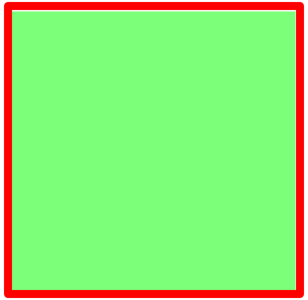
NUFFT With High Order Phase: 2^{nd} Order B_0



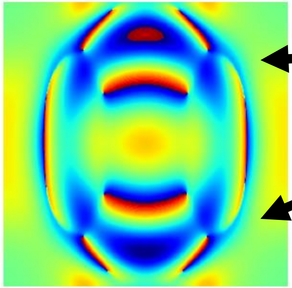
NUFFT With High Order Phase: 2^{nd} Order Strong B_0







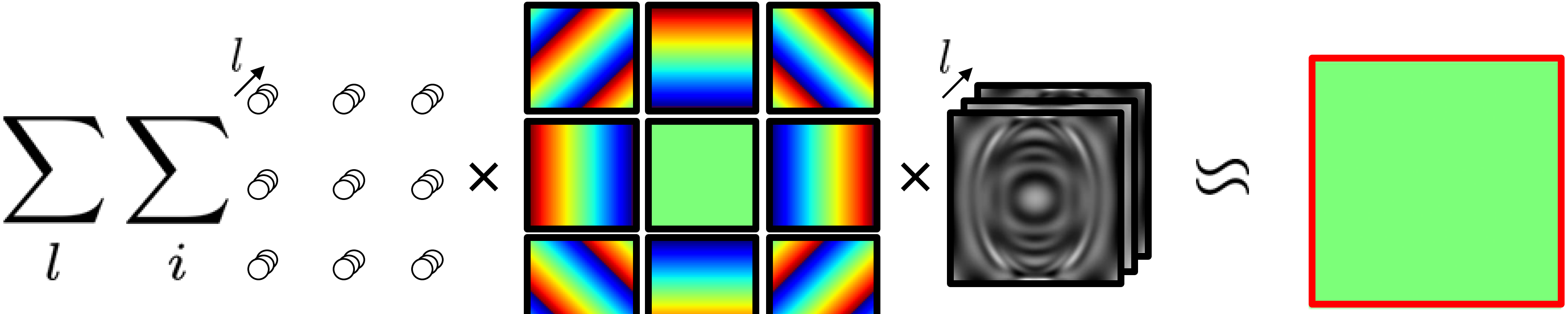
$$\underbrace{\sum_i \left[w_i(\boldsymbol{\alpha})\right] \times \left[e^{-j2\pi\mathbf{d}_i\cdot\mathbf{r}}\right] \times \left[b(\mathbf{r})\right]} \approx \left[e^{-j2\pi\phi(\mathbf{r})\cdot\boldsymbol{\alpha}}\right]$$



Struggles with phase wraps

Let's take inspiration from classical approaches and **add more spatial apodization functions**

HOFFT Model: Multiple Spatial Maps & Kernel Weights



$$\sum_l \sum_i \underbrace{[w_{l,i}(\boldsymbol{\alpha})] \times [e^{-j2\pi \mathbf{d}_i \cdot \mathbf{r}}] \times [b_l(\mathbf{r})]} \approx [e^{-j2\pi \boldsymbol{\phi}(\mathbf{r})\cdot\boldsymbol{\alpha}}]$$

Trades-off computation for accuracy

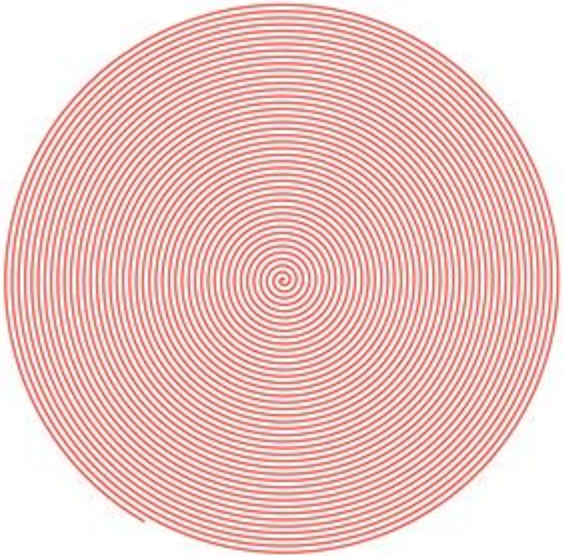
FFTs $\propto$ # Spatial Maps

Much better!

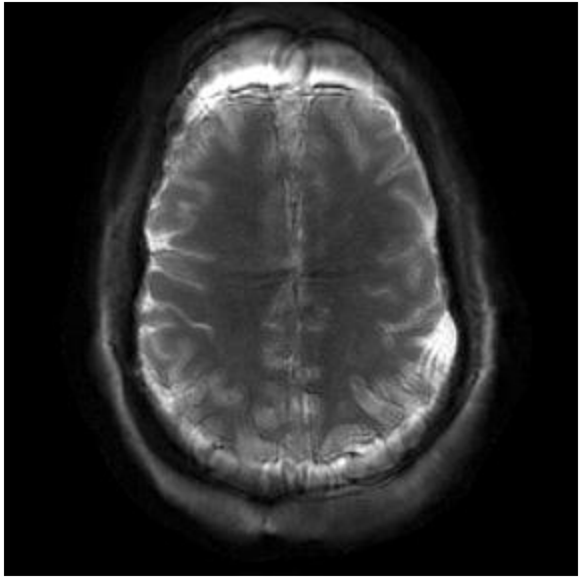
The **HOFFT** model has **multiple apodization functions** and corresponding **multiple kernels**

HOFFT Results

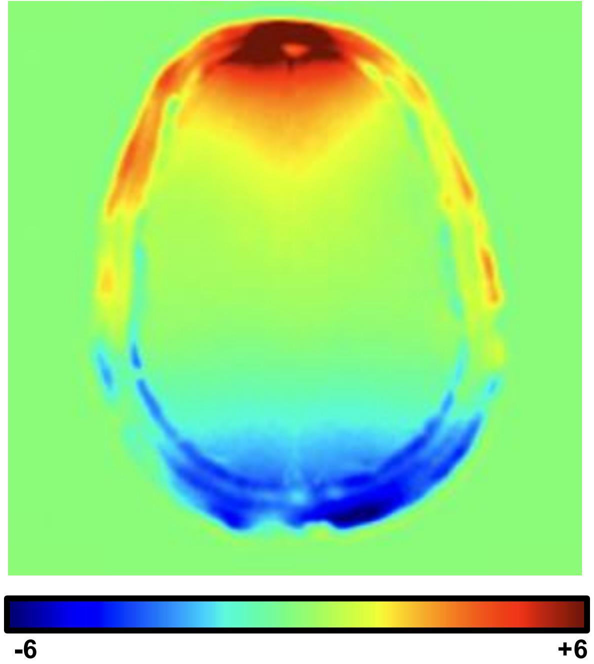
HOFFT For Physiological B0



R = 3 1mm Spiral



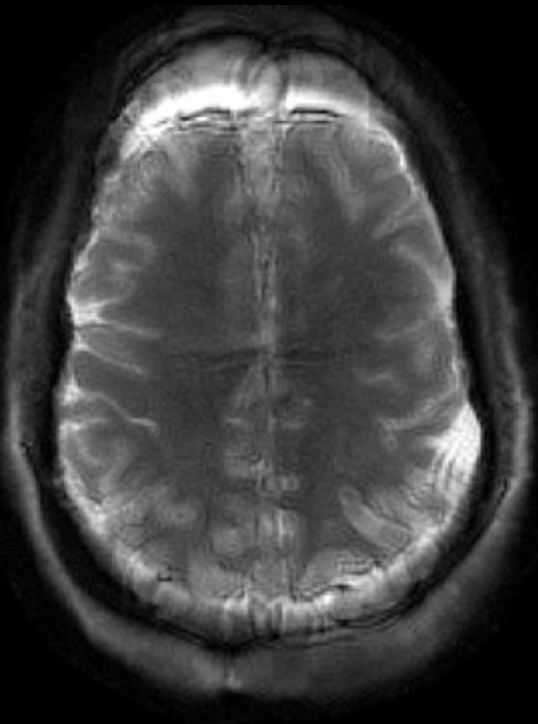
Uncorrected SENSE



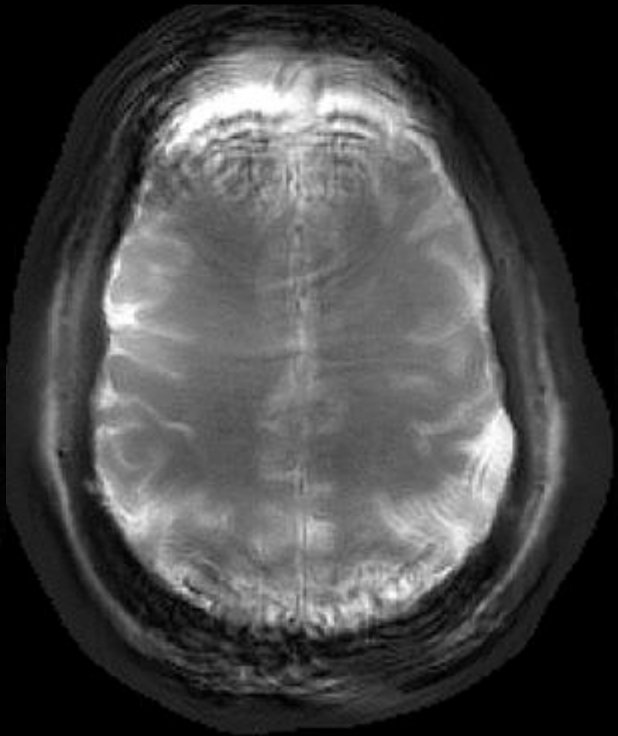
Cycles of B0 Phase
at End of Trajectory

*Acquired on a GE UHP 3T System + 32-channel NOVA coil

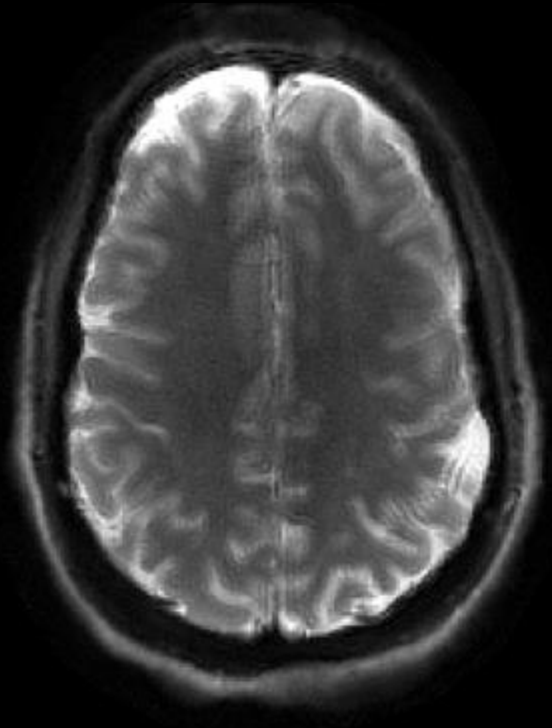
HOFFT For Physiological B0: Reconstructions



Uncorrected

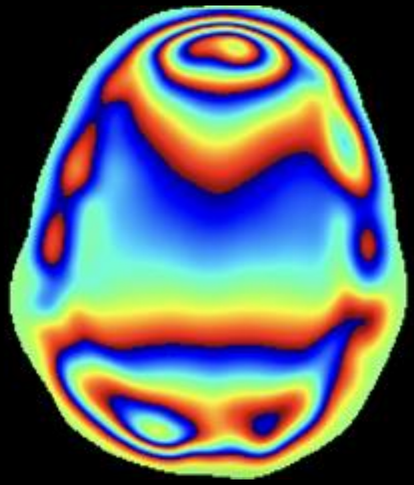


Rank-5 SVD + NUFFT

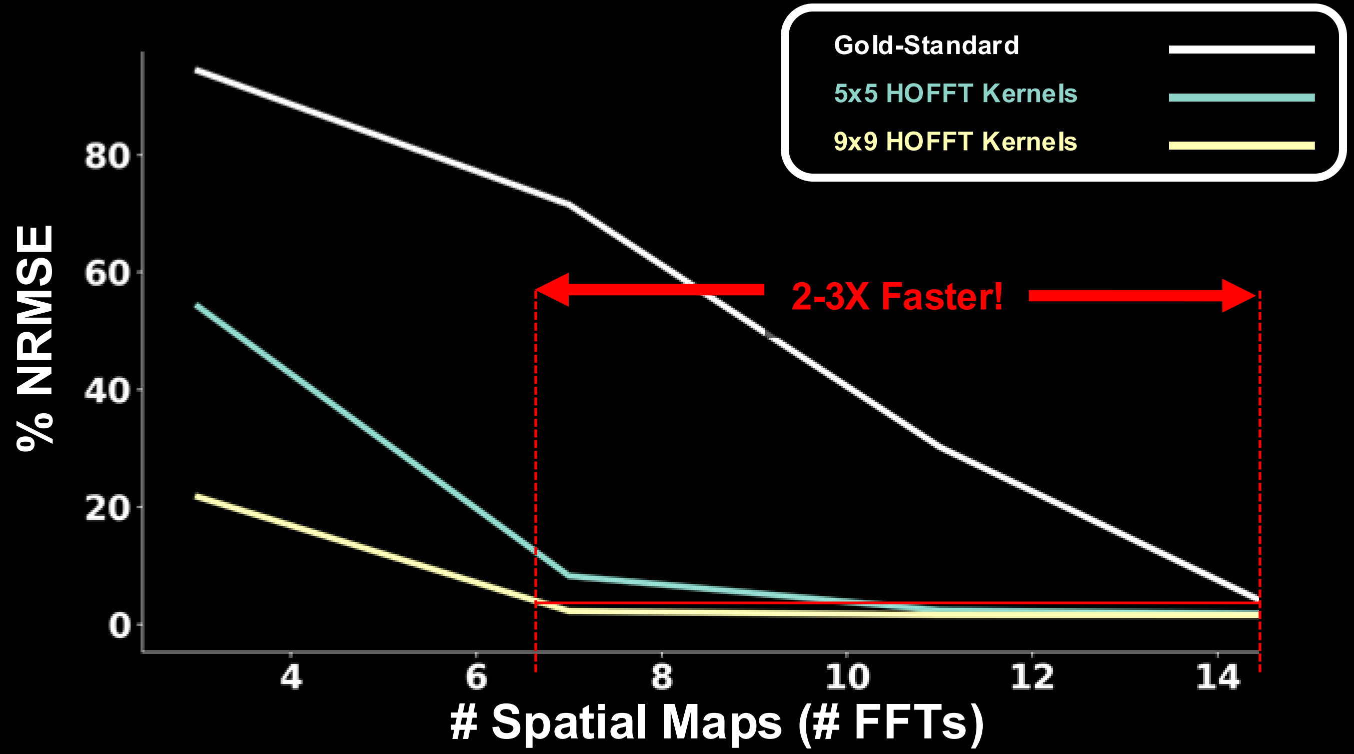


5 Spatial Maps + HOFFT

HOFFT For Physiological B0: Parameter Tradeoffs

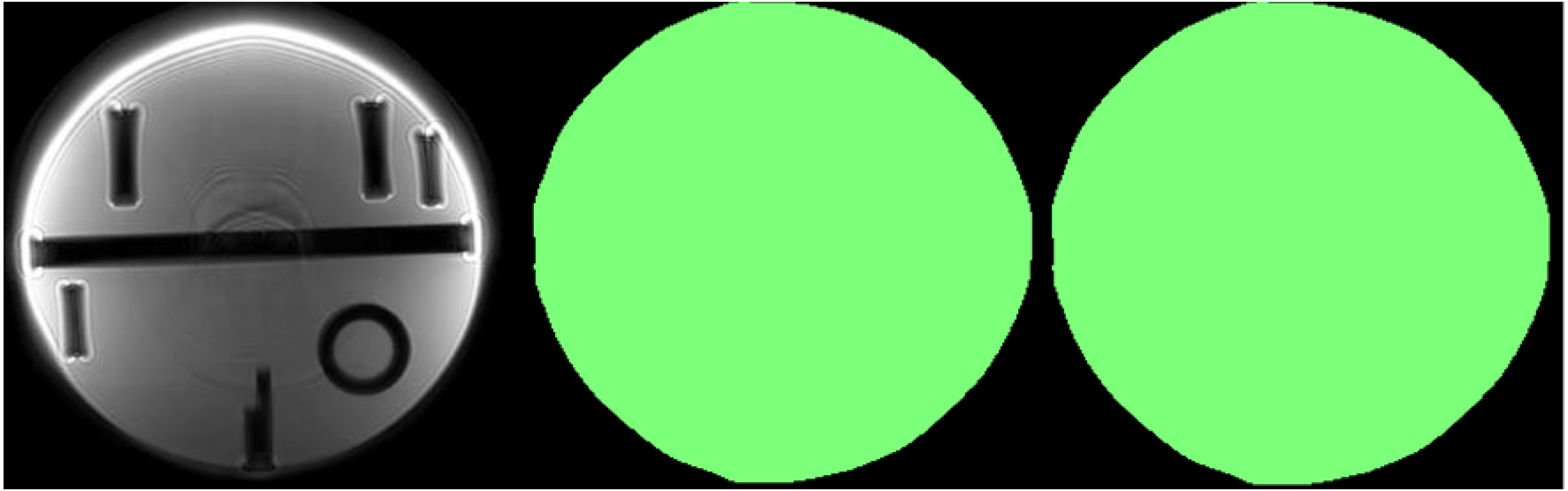


Physiological B0 Phase



HOFFT Coco & B0 Fields: Phase Evolution

R = 3 0.8mm Coronal Spiral



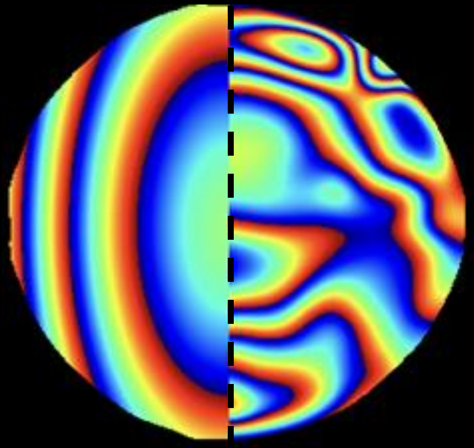
Uncorrected **Phase from B0** **Phase from Coco**

*Acquired on a GE UHP 3T System + 32-channel NOVA coil

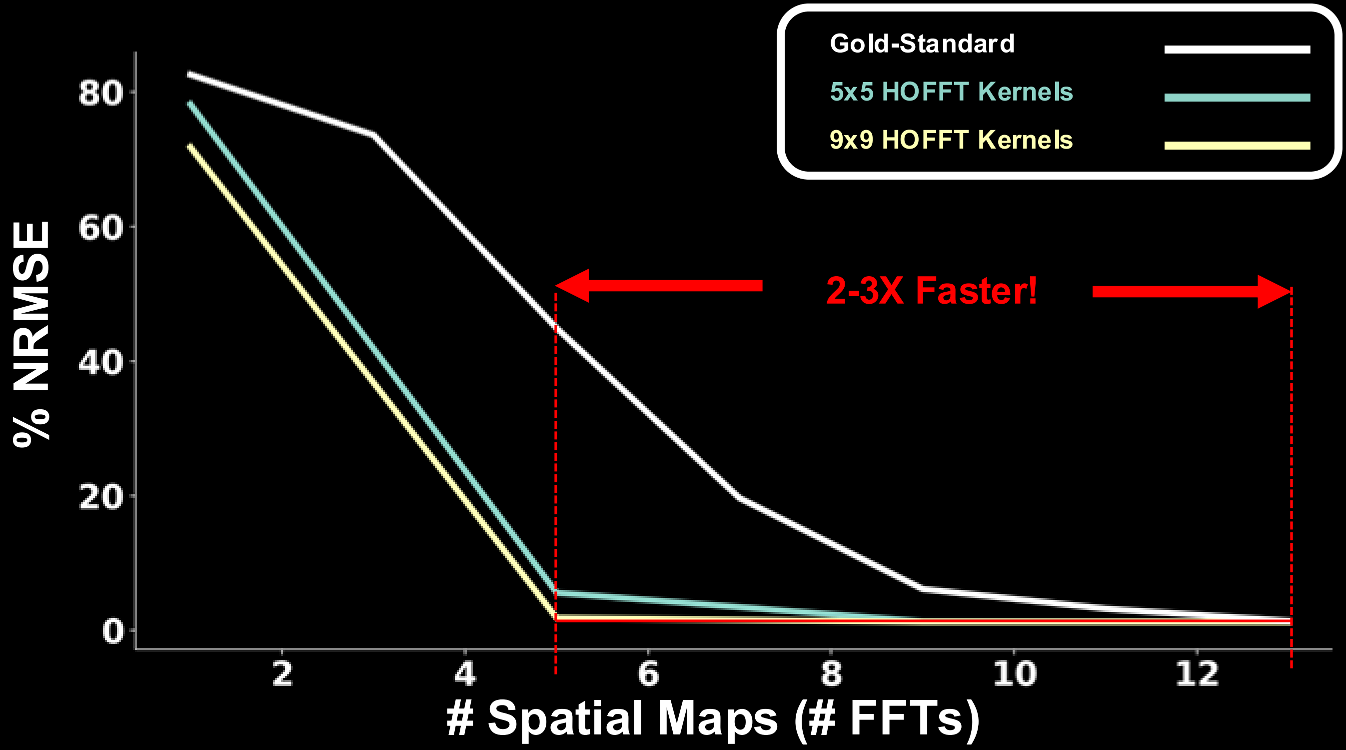
HOFFT Coco & B0 Fields: Reconstructions

Uncorrected | **Rank-5 SVD + NUFFT** | **5 Spatial Maps + HOFFT**

HOFFT Coco & B0 Fields: Parameter Tradeoffs



Concomitant & B0 Phase



Acknowledgments

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Setsompop Acquisition and Reconstruction Techniques